

1.8 The singular value decomposition and conditioning

The spectral theorem needs a symmetric matrix. A model matrix is rectangular, and the decomposition that fits it is the singular value decomposition.

Theorem 1.8.1 — Singular value decomposition

Let $\mathbf{A}$ be $m \times n$ of rank $r \geq 1$. There are an $m \times r$ matrix $\mathbf{U}$ and an $n \times r$ matrix $\mathbf{V}$, both with orthonormal columns, and numbers $\sigma_1 \geq \dots \geq \sigma_r > 0$, such that

$$\mathbf{A} = \mathbf{U}\mathbf{D}\mathbf{V}^\top = \sum_{i=1}^r \sigma_i \mathbf{u}_i \mathbf{v}_i^\top, \quad \mathbf{D} = \text{diag}(\sigma_1, \dots, \sigma_r). \quad (1.8.1)$$

The **singular values** σ_i are the positive square roots of the nonzero eigenvalues of $\mathbf{A}^\top \mathbf{A}$ (equivalently of $\mathbf{A}\mathbf{A}^\top$). The columns of $\mathbf{V}$ and $\mathbf{U}$ are eigenvectors of $\mathbf{A}^\top \mathbf{A}$ and $\mathbf{A}\mathbf{A}^\top$, with $\mathcal{C}(\mathbf{V}) = \mathcal{C}(\mathbf{A}^\top)$ and $\mathcal{C}(\mathbf{U}) = \mathcal{C}(\mathbf{A})$. Moreover $\mathbf{A}^+ = \mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$ is the Moore–Penrose inverse of $\mathbf{A}$.

Proof. $\mathbf{A}^\top \mathbf{A}$ is nonnegative definite of rank r (Proposition 1.2.3(c)). By the spectral theorem its eigenvalues are $\sigma_1^2 \geq \dots \geq \sigma_r^2 > 0$ and $n-r$ zeros, with orthonormal eigenvectors $\mathbf{V} = [\mathbf{v}_1, \dots, \mathbf{v}_r]$ for the nonzero ones and $\mathbf{V}_0$ for the zeros. Define $\mathbf{U} = \mathbf{A}\mathbf{V}\mathbf{D}^{-1}$. Then $\mathbf{U}^\top \mathbf{U} = \mathbf{D}^{-1} \mathbf{V}^\top \mathbf{A}^\top \mathbf{A} \mathbf{V} \mathbf{D}^{-1} = \mathbf{D}^{-1} \mathbf{D}^2 \mathbf{D}^{-1} = \mathbf{I}_r$. For each column $\mathbf{v}$ of $\mathbf{V}_0$, $\|\mathbf{A}\mathbf{v}\|^2 = \mathbf{v}^\top \mathbf{A}^\top \mathbf{A} \mathbf{v} = 0$, so $\mathbf{A}\mathbf{V}_0 = \mathbf{0}$. Since $\mathbf{V}\mathbf{V}^\top + \mathbf{V}_0\mathbf{V}_0^\top = \mathbf{I}$ (Proposition 1.5.3(d)),

$$\mathbf{A} = \mathbf{A}\mathbf{V}\mathbf{V}^\top + \mathbf{A}\mathbf{V}_0\mathbf{V}_0^\top = \mathbf{U}\mathbf{D}\mathbf{V}^\top.$$

Then $\mathbf{A}\mathbf{A}^\top \mathbf{U} = \mathbf{U}\mathbf{D}\mathbf{V}^\top \mathbf{V}\mathbf{D}\mathbf{U}^\top \mathbf{U} = \mathbf{U}\mathbf{D}^2$, so the columns of $\mathbf{U}$ are eigenvectors of $\mathbf{A}\mathbf{A}^\top$. $\mathcal{C}(\mathbf{U}) \subseteq \mathcal{C}(\mathbf{A})$ by definition of $\mathbf{U}$, and both have dimension r . The statement for $\mathbf{V}$ is the same argument applied to $\mathbf{A}^\top = \mathbf{V}\mathbf{D}\mathbf{U}^\top$. The four Penrose conditions for $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$ follow from $\mathbf{U}^\top \mathbf{U} = \mathbf{V}^\top \mathbf{V} = \mathbf{I}_r$. For instance, $\mathbf{A}\mathbf{A}^+ = \mathbf{U}\mathbf{U}^\top$ and $\mathbf{A}^+ \mathbf{A} = \mathbf{V}\mathbf{V}^\top$ are symmetric. ■

Completing $\mathbf{U}$ and $\mathbf{V}$ to orthogonal matrices gives the *full* SVD $\mathbf{A} = \tilde{\mathbf{U}}\tilde{\mathbf{D}}\tilde{\mathbf{V}}^\top$, with $\tilde{\mathbf{D}}$ of size $m \times n$ carrying $\sigma_1, \dots, \sigma_r$ on its diagonal and zeros elsewhere. By convention, $\sigma_{r+1} = \dots = \sigma_{\min(m,n)} = 0$. Geometrically, $\mathbf{A}$ maps the unit ball of $\mathbb{R}^n$ onto a solid ellipsoid in $\mathcal{C}(\mathbf{A})$ with semi-axes $\sigma_i \mathbf{u}_i$. The input direction $\mathbf{v}_i$ is stretched by the factor σ_i and turned into $\mathbf{u}_i$. The Moore–Penrose inverse undoes the stretching on $\mathcal{C}(\mathbf{A})$ and sends $\mathcal{C}(\mathbf{A})^\perp$ to zero. The matrix $\mathbf{U}\mathbf{U}^\top$ is the orthogonal projection onto $\mathcal{C}(\mathbf{A})$, whose theory is in [Chapter 6](#).

Proposition 1.8.2 — Norms and extremal properties

Let $\mathbf{A}$ be $m \times n$ with singular values $\sigma_1 \geq \sigma_2 \geq \dots$, and let $\|\mathbf{A}\|_2 = \max_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\| / \|\mathbf{x}\|$ be the **spectral norm**.

- (a) $\|\mathbf{A}\|_2 = \sigma_1$ and $\|\mathbf{A}\|_F^2 = \sum_i \sigma_i^2$.
- (b) If $\mathbf{A}$ has full column rank n , then $\min_{\mathbf{x} \neq \mathbf{0}} \|\mathbf{A}\mathbf{x}\|/\|\mathbf{x}\| = \sigma_n$, attained at $\mathbf{v}_n$.
- (c) (Eckart–Young.) For $k < r$, let $\mathbf{A}_k = \sum_{i \leq k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$. Then $\|\mathbf{A} - \mathbf{B}\|_2 \geq \sigma_{k+1} = \|\mathbf{A} - \mathbf{A}_k\|_2$ for every $\mathbf{B}$ with $\text{rank}(\mathbf{B}) \leq k$.

Proof.

- (a) and (b): $\|\mathbf{A}\mathbf{x}\|^2/\|\mathbf{x}\|^2 = \mathbf{x}^\top \mathbf{A}^\top \mathbf{A} \mathbf{x} / \mathbf{x}^\top \mathbf{x}$, and the eigenvalues of $\mathbf{A}^\top \mathbf{A}$ are the σ_i^2 (padded with zeros), so Theorem 1.7.7 applies. For the Frobenius norm, $\|\mathbf{A}\|_F^2 = \text{tr}(\mathbf{A}^\top \mathbf{A}) = \sum \sigma_i^2$ by Proposition 1.7.1(a).
- (b) $\mathbf{A} - \mathbf{A}_k = \sum_{i > k} \sigma_i \mathbf{u}_i \mathbf{v}_i^\top$ has largest singular value σ_{k+1} . If $\text{rank}(\mathbf{B}) \leq k$, then $\dim \mathcal{N}(\mathbf{B}) \geq n - k$, and since $(n - k) + (k + 1) > n$, a basis of $\mathcal{N}(\mathbf{B})$ together with $\mathbf{v}_1, \dots, \mathbf{v}_{k+1}$ is dependent. So the two subspaces share a unit vector $\mathbf{x} = \sum_{i \leq k+1} c_i \mathbf{v}_i$. Then $\|(\mathbf{A} - \mathbf{B})\mathbf{x}\|^2 = \|\mathbf{A}\mathbf{x}\|^2 = \sum_{i \leq k+1} \sigma_i^2 c_i^2 \geq \sigma_{k+1}^2$. ■

The Frobenius version of (c), $\|\mathbf{A} - \mathbf{B}\|_F^2 \geq \sum_{i > k} \sigma_i^2$, also holds (Eckart and Young 1936). It is the basis of principal components.

1.8.1 The condition number

Definition 1.8.1 — Condition number

The **condition number** of an $m \times n$ matrix $\mathbf{A}$ of full column rank is

$$\kappa(\mathbf{A}) = \frac{\sigma_{\max}(\mathbf{A})}{\sigma_{\min}(\mathbf{A})} = \frac{\sigma_1}{\sigma_n}.$$

For a nonsingular square matrix this equals $\|\mathbf{A}\|_2 \|\mathbf{A}^{-1}\|_2$. A matrix without full column rank has $\kappa(\mathbf{A}) = \infty$ by convention.

By Proposition 1.8.2, $\kappa(\mathbf{A})$ is the ratio of the largest to the smallest stretching that $\mathbf{A}$ applies to a unit vector. It is at least 1, is unchanged by scaling $\mathbf{A}$ or multiplying it on the left by a matrix with orthonormal columns, and equals 1 iff $\mathbf{A}^\top \mathbf{A} = c\mathbf{I}$, that is, iff the columns are orthogonal with equal lengths. Two further facts explain its role in regression. First, the singular values of $\mathbf{A}^\top \mathbf{A}$ are the σ_i^2 , so

$$\kappa(\mathbf{A}^\top \mathbf{A}) = \kappa(\mathbf{A})^2. \quad (1.8.2)$$

Second, κ bounds how much relative errors in the data can grow when a linear system is solved. If $\mathbf{A}$ is square and nonsingular, $\mathbf{A}\mathbf{x} = \mathbf{b}$ and $\mathbf{A}(\mathbf{x} + \delta\mathbf{x}) = \mathbf{b} + \delta\mathbf{b}$, then $\delta\mathbf{x} = \mathbf{A}^{-1}\delta\mathbf{b}$, and $\|\mathbf{b}\| \leq \sigma_1 \|\mathbf{x}\|$ gives

$$\frac{\|\delta\mathbf{x}\|}{\|\mathbf{x}\|} \leq \frac{\|\mathbf{A}^{-1}\|_2 \|\delta\mathbf{b}\|}{\|\mathbf{b}\|/\sigma_1} = \kappa(\mathbf{A}) \frac{\|\delta\mathbf{b}\|}{\|\mathbf{b}\|}. \quad (1.8.3)$$

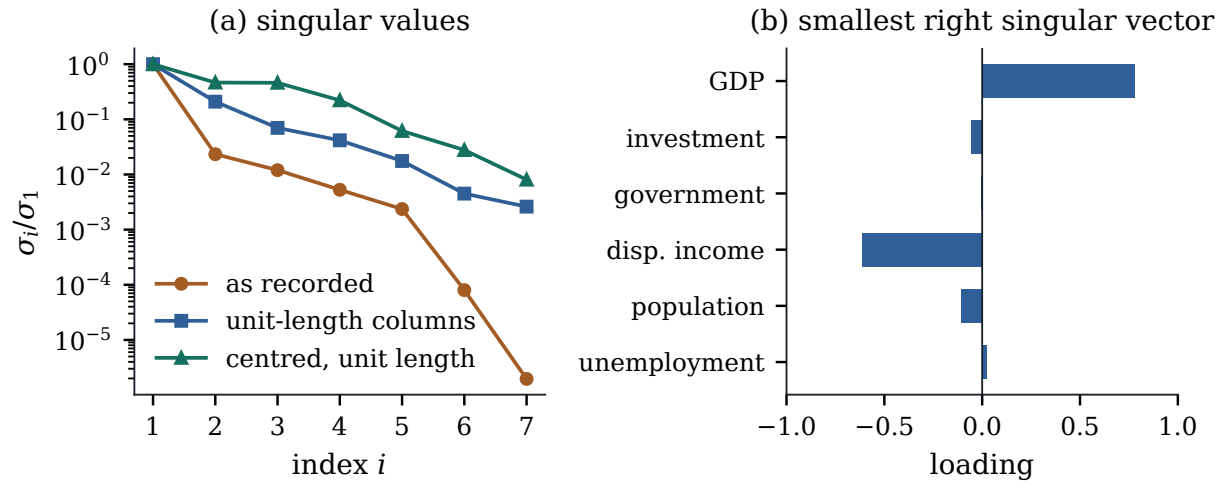


Figure 1.8.1: Singular value decomposition of a macroeconomic model matrix (Example (A macroeconomic design)). (a) Singular values relative to the largest, for the columns as recorded, rescaled to unit length, and centred then rescaled. The ratio of the first to the last is the condition number. (b) The right singular vector for the smallest singular value of the centred design. Its two large loadings reveal a near-dependence between GDP and disposable income.

In floating point, relative errors of about 10^{-16} in the data can therefore become errors of about $\kappa \times 10^{-16}$ in the solution. Solving the normal equations works with $\mathbf{X}^\top \mathbf{X}$ and so, by Eq. 1.8.2, squares the damage. Chapter 6 and Chapter 10 take this up.

A small $\sigma_{\min}$ has a direct statistical reading. By Proposition 1.8.2(b), $\|\mathbf{X}\mathbf{v}_n\| = \sigma_n$ with $\|\mathbf{v}_n\| = 1$. The coefficients $\mathbf{v}_n$ combine the columns of $\mathbf{X}$ into a vector that is nearly zero, which is a near-dependence among the regressors. This is the basis of the collinearity diagnostics of Chapter 26. The size of κ depends on the units of the columns, so columns are usually rescaled before it is interpreted.

Example. (A macroeconomic design)

Take the 203 quarterly US observations of real GDP, real investment, real government spending, real disposable income, population and unemployment, with an intercept, as a model matrix with $p = 7$ columns. As recorded, $\kappa(\mathbf{X}) = 5.10 \times 10^5$. Most of that is units: GDP is measured in billions and unemployment in percent. Rescaling each column to unit length reduces it to 383, and centring the regressors before rescaling reduces it further to 124. By Eq. 1.8.2 the corresponding cross-product matrix has condition number 15386.

What remains is not an artefact of units. Figure 1.8.1(b) shows the right singular vector for the smallest singular value, $\sigma_{\min} = 0.0175$, of the centred and scaled design. Its weight is concentrated on GDP (0.78) and disposable income (-0.61). The combination of the scaled columns with these coefficients is a vector of length 0.0175, although every column has length one. The two series move almost in lockstep: their correlation is 0.9990. No rescaling can separate their coefficients well. Centring is not a neutral choice, however: it removes near-dependences that involve the intercept, and Belsley et al. (1980) recommend scaling without centring for that reason, so their diagnostics would use the unit-length figure 383. Chapter 26 returns to the choice. The listing computes the singular values, and the script also checks Eq. 1.8.2, the Penrose conditions for $\mathbf{V}\mathbf{D}^{-1}\mathbf{U}^\top$, and the Eckart–Young bound.

```

import numpy as np
import statsmodels.api as sm
data = sm.datasets.macrodta.load_pandas().data
names = ["realgdp", "realinv", "realgovt", "realdpi", "pop", "unemp"]
labels = ["GDP", "investment", "government", "disp. income", "population", "unemployment"]
Z = data[names].to_numpy()
n = len(Z)

X = np.column_stack([np.ones(n), Z]) # as recorded
X_unit = X / np.linalg.norm(X, axis=0) # unit-length columns
Xc = np.column_stack([np.ones(n), Z - Z.mean(axis=0)]) # regressors centred
X_std = Xc / np.linalg.norm(Xc, axis=0) # ... then length 1

for name, M in [("raw", X), ("unit length", X_unit), ("centred", X_std)]:
    U, s, Vt = np.linalg.svd(M, full_matrices=False) # M = U diag(s) V'
    print(f"{name:12s} kappa = {s[0] / s[-1]:10.1f} singular values / s_1:",
          np.array2string(s / s[0], precision=4))

```

[Run this code online](#)

1.8.2 Exercises

1.8.2.1 B. Practice

Exercise B1.

Let $\mathbf{X} = [\mathbf{x}_1, \mathbf{x}_2]$ with $\|\mathbf{x}_1\| = \|\mathbf{x}_2\| = 1$ and $\mathbf{x}_1^\top \mathbf{x}_2 = r$, $|r| < 1$. Show that $\kappa(\mathbf{X}) = \sqrt{(1 + |r|)/(1 - |r|)}$. How large must $|r|$ be for $\kappa(\mathbf{X}) > 30$?

Solution. $\mathbf{X}^\top \mathbf{X} = \begin{pmatrix} 1 & r \\ r & 1 \end{pmatrix}$ has eigenvalues $1 \pm r$, with eigenvectors $(1, \pm 1)^\top$. The singular values of $\mathbf{X}$ are their square roots, so $\kappa(\mathbf{X})^2 = (1 + |r|)/(1 - |r|)$. For $\kappa > 30$ we need $(1 + |r|)/(1 - |r|) > 900$, that is, $|r| > 899/901 \approx 0.998$. For two columns, even $\kappa(\mathbf{X}) = 100$ requires $|r| > 9999/10001 \approx 0.9998$.

Exercise B2.

Show that $\kappa(\mathbf{X}) \geq \max_j \|\mathbf{x}_j\| / \min_k \|\mathbf{x}_k\|$, so that columns on very different scales force a large condition number. Show by example that rescaling the columns can also increase κ .